

Change in Single People's Attachment and Co-Development with Life Satisfaction

– Preregistration –

Based on their experiences in close relationships, people form internal working models of attachment (Bowlby, 1980; Hazan & Shaver, 1987), which comprise both global and relationship-specific components (e.g., attachment models of a partner or friends; Fraley et al., 2011; Klohnen et al., 2005; Overall et al., 2003). Securely attached individuals (i.e., those low in attachment avoidance and anxiety) typically exhibit more favorable relationship outcomes, well-being, and self-esteem (Choi et al., 2026; Joel et al., 2020; Mikulincer & Shaver, 2012, 2019). Although attachment research has traditionally focused on romantic relationships (e.g., Hazan & Shaver, 1987; Heffernan et al., 2012; Lehnart & Neyer, 2006; Traut, Gander, Uhlich, et al., 2025), an emerging body of work has begun to investigate attachment among single people (Brumbaugh, 2017; Oh et al., 2026; Pepping et al., 2018; Schachner et al., 2008). Because long-term single people differ in their organization of close relationships from partnered individuals, attachment processes identified in predominantly partnered samples may not generalize to this population.

Recent studies suggest that anxious or avoidant attachment is linked to poorer well-being and relationship outcomes in many long-term single people (MacDonald & Park, 2022; Oh et al., 2026; Park et al., 2023; Pepping et al., 2025; Walsh et al., 2024). There are, however, three major limitations of existing research that constrain our understanding of attachment among single people. First, most of this research has investigated general attachment orientation, not differentiating relationship-specific attachment models. This narrow focus is suboptimal from a measurement perspective because some items in common attachment scales explicitly refer to romantic partners. Therefore, such general scales blend romantic and global attachment (cf. MacDonald & Park, 2022).¹ In addition, this practice leaves open the question how single people experience attachment in other, non-romantic relationships. In lieu of romantic relationship experiences, single people likely experience attachment-relevant situations through their interpersonal relations with friends or family members, although their perceived social support is on average lower than that of partnered individuals (Girme et al., 2022; Stronge et al., 2019).

¹ Note that we use *general* attachment to refer to studies using undifferentiated measures that may confound global and romantic attachment models. We use *global* attachment for research based on the ECR-RS measure (Fraley et al., 2011) that explicitly differentiates between domains of attachment models.

Second, most previous research on attachment in single people has been cross-sectional (cf. Traut, Gander, Weidmann, et al., 2025), and little is known about how attachment changes in single people—as they age or with increasing singlehood duration. This constrains theoretical progress regarding why single individuals, on average, report lower attachment security than partnered individuals. These differences may be the result of selection and socialization processes among both single and partnered people. Among long-term single people, attachment may remain relatively stable over time because they do not experience the socialization effects of committed romantic relationships or because they form stable non-romantic attachment relationships. Alternatively, ongoing singlehood, especially if seen as an undesired state, may contribute to maladaptive changes in attachment security over time.

Third, in addition to the developmental course of attachment in single people, its co-development with well-being remains unknown. Prior research has documented mean-level decreases in well-being among long-term single people compared to partnered young adults (Krämer et al., 2026). In young adult single people, potentially maladaptive developmental trajectories of attachment may undermine the maintenance of existing relationships or formation of new ones and be costly for well-being over time (Girme et al., 2022; Kansky, 2018; Perelli-Harris et al., 2019). This would be in line with theoretical accounts of functioning romantic relationships as developmental tasks (Hutteman et al., 2014; Lodi-Smith & Roberts, 2007). Still, some single people find secure attachment in other close relationships and thereby experience beneficial well-being change. This again underscores the need to investigate different attachment domains among single individuals.

Here, we aim to close several gaps in knowledge by investigating how single people change in different attachment models (i.e., global, partner, best friends, parents) over the course of long-term singlehood compared to partnered people. We compare consistently single people to individuals in stable romantic relationships but also to individuals who are intermittently partnered/single. Further, we plan on examining how singles' attachment and life satisfaction co-develop with increasing singlehood duration. A better understanding of attachment in singlehood and its development is particularly relevant given the rising prevalence of singlehood across many countries (Adamczyk & Trepanowski, 2023; Esteve et al., 2020; Pepping et al., 2018), and emerging evidence of cohort declines in attachment security (Konrath et al., 2014). As a result,

social scientists are increasingly exploring the implications of singlehood across the lifespan (Beckmeyer & Jamison, 2023; Girme et al., 2023).

Singlehood in Young Adulthood

Young adults who stay single for longer periods of time may face challenges for their well-being and psychosocial adjustment. Long-term romantic relationships have long been regarded as the societal norm (DePaulo & Morris, 2006; Soons et al., 2009), and are frequently associated with greater well-being and health in large, representative samples (e.g., Braithwaite et al., 2010; Gonzalez Avilés, 2024; Krämer et al., 2025; Oh et al., 2022; Purol et al., 2021; Roelfs et al., 2011; Stern et al., 2024; Uunk & Hoffmann, 2022). A recent study focusing on long-term singles' well-being in young adulthood found that those who remained single (from ages 17 to 29) showed greater decreases in life satisfaction and increases in loneliness compared to those who eventually found a partner (Krämer et al., 2026). These average well-being deficits were minimal in adolescence but became more pronounced in the late 20s, when depressivity increases (i.e., in self-reported symptoms) also diverged between consistently single people and those who eventually partnered. Lower momentary well-being also decreased the odds of eventually finding a partner (Krämer et al., 2026), similar to a negative feedback loop. Conversely, experiencing a first romantic relationship reversed adverse changes in life satisfaction and loneliness (but not depressivity) to some extent.

Different theoretical perspectives offer explanations for these maladaptive changes in well-being. Developmental task theories suggest that missing the important task of relationship formation in young adulthood has downstream consequences (Havighurst, 1972; Hutteman et al., 2014; Lodi-Smith & Roberts, 2007). Compared to young adults with partnership experiences, single people may have fewer opportunities to acquire positive close relationship experiences (see also Arriaga et al., 2018). These close relationship experiences may, in turn, contribute to identity formation, emotional resilience, interpersonal competence, and secure attachment during the transition to established adulthood (Barzeva et al., 2021; W. A. Collins et al., 2009; Gonzalez Avilés et al., 2021, 2023; Traut, Gander, Weidmann, et al., 2025). In the latter phase of emerging adulthood, as the desire for a committed romantic relationship increases on average (Arnett, 2000; Shulman & Connolly, 2013), some young adults may fall behind in the sense that they are less likely to form and maintain stable intimate relationships after longer singlehood spells.

Even though many theoretical approaches tend to frame singlehood as a deficit—portraying single people as unhappy or unable to find a partner—other perspectives highlight potential long-term advantages of singlehood, including a greater focus on education, career, social life, and work–life balance (e.g., DePaulo, 2023; Kislev, 2019; Park et al., 2023; van den Berg & Verbakel, 2022). For some, singlehood may provide autonomy, individuality, and psychological richness (Kislev, 2018; Oishi & Westgate, 2022; Ryan & Deci, 2000), while also allowing for strong social commitments to nonromantic relationships (Burton-Chellew & Dunbar, 2015). Mean-level differences in well-being between consistently single people and partnered individuals may originate from social pressure and stigma towards singles. In addition, group comparisons mask heterogeneity among both groups. For example, subgroups of single young adults are happy with their situation, including but not limited to aromantic people who do not express pronounced desire for a romantic relationship (Canaletti et al., 2026; Carvalho & Rodrigues, 2022). However, initial research indicates that there are many single people who are unhappy and dissatisfied with being single (Lehmann et al., 2015; Park et al., 2023; Pepping et al., 2025), which may especially be the case in earlier compared to later phases of the lifespan (Oh et al., 2022; Park et al., 2022).

One central individual difference that may explain these heterogeneous experiences (e.g., well-being differences) and perspectives on singlehood is attachment style, which has received increased attention in recent research on singlehood and partnership transitions (e.g., Pepping et al., 2025; Traut, Gander, Weidmann, et al., 2025). Importantly, even if single people gain fewer attachment-relevant experiences in partnerships, they might still feel securely attached with nonromantic close others such as friends or parents. However, most research on single people has focused on general attachment, neglecting the distinction of different attachment working models for between-person individual differences as well as for longitudinal attachment development and co-development with well-being.

Attachment Perspectives on Singlehood

Attachment theory describes how people regulate their emotions through ties to close others and the implications of these socio-emotional regulation tendencies for how relationships are formed and maintained (Fraley, 2019; Fraley & Shaver, 2000; Mikulincer & Shaver, 2012). The theory originated from observations of infant–caregiver emotional bonds (Bowlby, 1979, 1980) and was subsequently extended to adult romantic relationships (Hazan & Shaver, 1987;

Shaver et al., 1988; Shaver & Hazan, 1988) as well as other important pair-bonded social relationships (with family, friends, and other social network partners; Gillath et al., 2017). Attachment working models continue to shape close relationships across the lifespan (Pietromonaco & Barrett, 2000). Individuals with secure attachment models typically feel worthy of love and comfortable seeking emotional support, whereas those with insecure attachment (i.e., high in avoidance and/or anxiety) exhibit lower self-worth and interpersonal trust and experience more conflictual and less satisfying relationships (Brassard et al., 2009; N. L. Collins & Feeney, 2004). Specifically, attachment anxiety is characterized by heightened sensitivity to rejection, excessive reassurance-seeking, and emotional distress (Fraley, 2019). Attachment avoidance, in contrast, is marked by a preference for self-reliance, discomfort with emotional closeness and disclosure, and lower inclination toward intimacy. Attachment models comprise global and relationship-specific components (Klohn et al., 2005; Overall et al., 2003; Pierce & Lydon, 2001). Although there is some consistency in attachment models from one relationship to the next and across relationship types (Brumbaugh & Fraley, 2006, 2007), attachment models for one's parents, friends, and romantic partners also have unique components (Dugan et al., 2022).

Attachment shows moderate rank-order stability in adulthood (Chopik et al., 2024). In addition to rank-order stability, some studies have examined age-related differences (in cross-sectional data) and mean-level change in attachment across age (in longitudinal data). Cross-sectionally, individuals report higher attachment avoidance with older age across different relationship types and lower attachment anxiety specifically with romantic partners and friends (Chopik et al., 2013; Chopik & Edelstein, 2014; Hudson et al., 2015). Longitudinal research, however, also indicates mean-level decreases in global attachment avoidance across the lifespan, though these are less pronounced than decreases in anxiety (Chopik et al., 2019; Dugan et al., 2022; Traut, Gander, Uhlich, et al., 2025). Evidence also supports co-development among different types of attachment working models. Short-term perturbations in one working model (e.g., with a romantic partner) may influence other models (e.g., parent-specific or global attachment) over time, representing transactions across the trait- and state-levels of attachment models (Lee & Gillath, 2016). According to one study, people who developed greater security in their relationships with their parents, partners, or best friends over time also experienced an overall increase in global attachment security (Dugan et al., 2022). Such co-development was

most pronounced between global attachment models and those of a romantic partner, and less so for friend-specific attachment.

Building on attachment theory, several accounts explain evidence that links being single to lower general attachment security compared to being in a relationship (Chopik et al., 2013, 2019; Schachner et al., 2008; Traut et al., 2026; Traut, Gander, Weidmann, et al., 2025). The Attachment Security Enhancement Model (Arriaga et al., 2018) proposes that secure working models of attachment are fostered through repeated experiences in security-enhancing relationship situations. At least in the domain of partnered relationships, younger single people may lack exposure to security-enhancing situations compared to those who find a partner. For some individuals, failing to find a committed romantic partner may be maladaptive for attachment models, particularly in the romantic domain, through repeated experiences of perceived rejection or abandonment. These maladaptive effects for partner working models may also spill over to global attachment models (Dugan et al., 2022). At the same time, global attachment models are also shaped by the quality of close non-romantic relationships, such as friendships and family bonds, which provide alternative sources of attachment-relevant experiences for long-term single people.

Accordingly, theoretical perspectives on attachment and long-term singlehood emphasize diversity in singlehood experiences (see also Girme et al., 2023) and propose distinct pathways linking attachment to being and staying single (Pepping et al., 2018; Pepping & MacDonald, 2019). Some single people are theorized to remain single because their attachment system in the romantic domain is less active (e.g., avoidant of closeness, commitment, and intimacy) or hyperactive (i.e., anxious and sensitive to rejection and abandonment; see also Mikulincer & Shaver, 2012) compared to partnered people in happy relationships. Other single people are securely attached (e.g., to friends and other family; Brumbaugh, 2017), have low to moderate desire for a romantic relationship, and generally report higher well-being than involuntarily single people (MacDonald & Park, 2022; Oh et al., 2026). Thus, singlehood clearly also reflects a personal choice for some people (DePaulo, 2023; Lehmann et al., 2015; Pepping et al., 2018). In such cases, voluntary long-term singlehood is experienced mostly positively (Park et al., 2023) and may be unrelated to maladaptive trajectories. The idea of different pathways into singlehood has found initial support such that single adults with lower general attachment

security report more maladaptive outcomes (Pepping et al., 2025), similarly for attachment anxiety and avoidance (MacDonald & Park, 2022).

The Importance of Relationship-Specific Attachment

Beyond general or global attachment, however, evidence is still lacking on singles' specific attachment models. The distinction between global and relationship-specific attachment is particularly relevant for research on long-term single people because their attachment-relevant experiences may be rooted primarily in friendship and family relationships. Research indicates that non-romantic relationships are a common source of satisfaction with singlehood (Spielmann et al., 2013; Walsh et al., 2022, 2024). One study also suggests that long-term single people maintain more network ties to friends and family than married individuals (Sarkisian & Gerstel, 2016). Thus, it is crucial to distinguish between global and relationship-specific attachment working models to provide a more comprehensive theoretical understanding of why some single individuals thrive whereas others do not.

Besides these conceptual issues, common measurement practices have obscured what we know about attachment in single people. Established attachment measures (e.g., the ECR or ECR-R; Brennan et al., 1998; Fraley et al., 2000) confound global and romantic attachment models, because two types of items form common scores: (1) items that apply to close relationships in general (e.g., "My desire to be very close sometimes scares people away") and (2) items that specifically refer to romantic partners (e.g., "I tell my partner just about everything"). This is not necessarily an issue if researchers are working with partnered samples and are interested in general attachment. However, responding to items that assume a romantic partner poses challenges for consistent or long-term single people. Stated differently, traditional measures of general attachment are not as general as they may seem at first glance; they contain items that are about the self, about others in general, about specific partners, and about romantic relationships broadly speaking. A domain-focused, structural measure (i.e., the ECR-RS; Fraley et al., 2011) is necessary for singlehood research, because it allows for a differentiation of attachment to different types of relationships.

Given these measurement limitations, there is little evidence on how single people differ from partnered individuals in their global and relationship-specific attachment models. Previous evidence using general attachment measures indicates that, on average, single people have lower general attachment security than partnered people (Chopik et al., 2013, 2019; Schachner et al.,

2008; Schindler et al., 2010). Recent studies using the global (Traut et al., 2026; Traut, Gander, Weidmann, et al., 2025) and romantic² (Weidmann et al., 2024) subscales of the ECR-RS corroborated these findings, also indicating less secure attachment in single people. So far, no studies have investigated differences in best-friend-, mother-, and father-specific attachment between single and partnered people. Perspectives on thriving singlehood suggest that single people form securely attached friendships that fulfill some of the functions of romantic relationships in terms of emotional closeness and need satisfaction (Burton-Chellew & Dunbar, 2015; DePaulo, 2023; Sarkisian & Gerstel, 2016; Walsh et al., 2024). Conversely, one might argue that some spillover from global to domain-specific attachment is likely (Dugan et al., 2022), and that insecure global attachment seems to select people into singlehood, relating to lower odds of finding a partner and higher odds of separation (Fraley et al., 2021; Kirkpatrick & Hazan, 1994; Peters et al., 2025; Schindler et al., 2010; Traut, Gander, Weidmann, et al., 2025). Overall perceived social support also tends to be lower in single compared to partnered people (Stronge et al., 2019). Therefore, comparatively insecure domain-specific attachment in single people also remains a possibility and warrants further investigation.

The Roles of Selection and Socialization for Single and Partnered Individuals' Attachment

So far, however, the developmental processes through which *long-term single people's* attachment models are maintained or changed over time remain understudied. One study combined several smaller longitudinal samples and found that being single was associated with higher general attachment anxiety and avoidance throughout adulthood (Chopik et al., 2019). Age-related changes followed reverse U-shape curves that were mostly parallel between single and partnered individuals, except for greater steepness in single people's late-life anxiety decreases. However, this study was limited by rather small sample sizes, infrequent assessments, and undifferentiated measurement of general but not relationship-specific attachment (based on a California Adult Q-Sort measure).

To systematically characterize transactions of attachment and singlehood over time, a personality development lens of selection and socialization processes can be applied (Roberts, 2018; Roberts & Nickel, 2017). Selection effects imply that individual differences in attachment may influence who enters and maintains stable romantic relationships over time. Attachment

² Attachment assessments using the ECR-RS romantic subscale in single people typically refer to past and hypothetical romantic partners (Fraley et al., 2021; Lehnart & Neyer, 2006; Weidmann et al., 2024).

models are associated not only with experiences and behaviors within social relationships but also with their formation. Higher global and romantic partner attachment avoidance relate to a lower likelihood of subsequent partnering (i.e., prospective selection effects; Schindler et al., 2010; Traut, Gander, Weidmann, et al., 2025), whereas higher global attachment anxiety and avoidance relate to a higher likelihood of experiencing relationship separation (Fraley et al., 2021; Kirkpatrick & Hazan, 1994; Peters et al., 2025). These findings suggest that attachment insecurity may affect selection into long-term singlehood, such that individuals who remain single exhibit lower attachment security than those who eventually enter a relationship. However, this prediction may not apply to all long-term single individuals. Many single people report low to medium relationship desire or voluntarily prioritize other life domains and goals (Adamczyk, 2021; Girme et al., 2023; Pepping et al., 2025). Thus, an alternative prediction applies if long-term singlehood reflects lower motivation to pursue romantic relationships rather than relationship difficulties. In this case, attachment security may not systematically differ between individuals who remain single and those who subsequently form romantic partnerships.

In addition to differential selection, socialization effects may explain attachment differences between single and partnered individuals. For this comparison, socialization processes can be viewed from two perspectives based on relationship status: (1) experiences by (eventually) partnered individuals that long-term single people miss out on, and (2) experiences single people tend to make but partnered people do not.

First, romantic relationship events (e.g., formation, break-up, parenthood) and qualitative relationship indicators such as conflict and support have been shown to shape attachment models in adulthood (Fraley et al., 2021; for an overview, see Fraley & Hudson, 2017). Specifically, both romantic relationship formation and separation relate to subsequent changes in attachment (Davila & Sargent, 2003; Fraley et al., 2021; Ruvolo et al., 2001; Traut, Gander, Weidmann, et al., 2025). While relationship formation is followed by short-term decreases in attachment anxiety and avoidance (i.e., global and romantic attachment; Chopik et al., 2024; Fraley et al., 2021), break-ups are associated with increases in attachment avoidance (Traut, Gander, Weidmann, et al., 2025) and, when the individual is the recipient rather than initiator of the break-up, increases in attachment anxiety (Spielmann et al., 2016). Together, this is consistent with the Attachment Security Enhancement Model (Arriaga et al., 2018) and developmental task

perspectives³. Long-term single people may lack exposure to important age-normative social role transitions (i.e., formation of a stable romantic relationship) that provide security-enhancing experiences. For example, positive, intimacy-related relationship experiences are associated with further decreases in attachment avoidance (Bayraktaroglu et al., 2023; Stanton et al., 2017). Over time, satisfying romantic relationship experiences may cause partnered individuals to differ in attachment orientations from long-term single people. Based on theoretical notions and prior knowledge, we may expect that these socialization processes primarily affect romantic attachment, with lagged spillover to global attachment change (Dugan et al., 2022). At the same time, knowledge on friend- or parent-specific attachment is highly limited, and the degree of domain spillover in this specific context unknown. Further, whether individuals with romantic relationship experiences change towards greater attachment security or insecurity likely depends on the stability and quality of their relationship(s). Break-ups have been shown to reduce attachment security (Fraley et al., 2021; Traut, Gander, Weidmann, et al., 2025). Therefore, individuals in stable long-term relationships may develop towards greater romantic and global attachment security, whereas those who experience both relationship formations and break-ups intermittently do not.

Second, socialization processes that are specific to singlehood (i.e., experiences that partnered people do not make) may explain differences in attachment, although much less evidence is available compared to research on relationships. For some people, long-term singlehood reflects an undesired state that is accompanied by dissatisfaction of closeness and intimacy needs (Gonzalez Avilés, 2024; Hofer & Hagemeyer, 2018; Ryan & Deci, 2000), and is incompatible with important life goals of relationship and family formation (Buchinger et al., 2022). Such experiences of singlehood are likely characterized by perceived rejection or exclusion, which may lead people to shift their standards in terms of partner preferences (Spielmann et al., 2013) or disengage from seeking romantic relationships (Wrosch & Heckhausen, 1999). In addition, even in single people with comparatively less interest in a romantic relationship, social stigma with increasing singlehood duration and age may reinforce

³ In line with ideas of normative developmental timing, the socialization effects of relationship experiences may vary across lifespan phases and the first serious romantic relationship experiences in young adulthood may be particularly pivotal for attachment stability or change later in life (esp. for romantic-partner attachment). This is captured by the concept of canalization (Dugan et al., 2024; Fraley & Hudson, 2017): experiences in early adulthood, when attachment stability is lower, tend to have more lasting and formative effects than those in middle or older adulthood (Chopik et al., 2024).

negative self-perceptions (Fisher & Sakaluk, 2020; Girme et al., 2022; Greitemeyer, 2009; Sprecher & Felmlee, 2021), which could also translate into maladaptive attachment change. Together, singlehood—especially if perceived as involuntary—may increase global and romantic attachment insecurity over time spent in singlehood. This may occur independently of selection effects (i.e., single and eventually partnered individuals starting at similar initial levels) or in conjunction with them. Joint selection and socialization effects would be more consistent with prior evidence on global attachment (see above).

At the same time, more adaptive sides of long-term singlehood for attachment change may have been obscured by measurement practices that conflate global and romantic attachment. At least cross-sectionally, there is evidence that some long-term single people are securely attached to close others in non-romantic relationships (MacDonald & Park, 2022; Pepping et al., 2025; Schachner et al., 2008). Examining non-romantic attachment may complement the “missed event” perspective with a more adaptive developmental outlook. Specifically, friend-specific attachment orientations may develop towards greater security in long-term single people, with some evidence suggesting that, compared to partnered people, they focus more or as much on establishing networks of emotionally close and supportive non-romantic relationships (Adamczyk, 2016; Girme et al., 2023; Sarkisian & Gerstel, 2016; cf. Stronge et al., 2019).

Altogether, the comparison of attachment trajectories between long-term single people and partnered individuals remains incomplete (see also Gonzalez Avilés, 2024), as average differences in attachment security between long-term single people and partnered individuals may be accounted for by selection as well as socialization links between attachment and experiences either in relationships or in singlehood. A recent study found baseline differences in global attachment anxiety between single and partnered young adults, which narrowed over a 2.5-year period for initially single people after partnering (Traut, Gander, Weidmann, et al., 2025), though findings for avoidance were less clear. Based on the (sparse) existing evidence, we expect that consistently single people show less adaptive development in global and romantic attachment than consistently partnered individuals. However, we acknowledge that alternative pathways are also theoretically plausible, including initial selection effects followed by relative stability in single people’s attachment. In addition, the other attachment domains (i.e., friends and parents, which we examine exploratorily) may differ from this hypothesized pattern for global and romantic attachment.

Attachment–Well-Being Co-Development Within Singlehood

Aside from attachment-related differences in the selection and maintenance of relationships or singlehood, an increasingly important question is whether single people are happy overall and which personal and relational characteristics predict their well-being (Girme et al., 2023). Across several indicators of well-being, recent research has documented maladaptive mean-level trajectories among long-term single young adults (Gonzalez Avilés, 2024; Gonzalez Avilés et al., 2021). Here, we focus on life satisfaction as the cognitive-evaluative component of subjective well-being (Diener et al., 2018; Luhmann et al., 2021). On average, life satisfaction declines during young adulthood among consistently single people relative to individuals who form a romantic partnership, with these declines becoming more pronounced in the late twenties (Krämer et al., 2026). However, beyond these mean-level changes in well-being, it remains unknown how single individuals' well-being develops jointly with attachment in different domains. In other words, heterogeneity in attachment levels and change may be linked to differential well-being trajectories. By investigating such co-development, we extend singlehood research towards a broader and more qualitative account of single people's social relationship experiences.

Even if long-term single people, on average, report lower well-being and greater attachment insecurity, substantial heterogeneity exists within this population (Girme et al., 2023; Park et al., 2022; Pepping et al., 2025). Some single people may be securely attached in high-quality non-romantic relationship, experience high levels of well-being, and have little desire to enter a romantic relationship. This pattern may be particularly evident for attachment to friends, who can serve as important sources of support, intimacy, and belonging in the absence of a romantic partner (Fisher et al., 2021). Thus, beyond the absence of attachment-relevant romantic experiences (Arriaga et al., 2018), strong attachment security in non-romantic domains may provide alternative attachment-relevant sources of support and security. This may promote positive developmental trajectories especially in friend-specific attachment and spill over to more secure global attachment and greater well-being, too. Examining co-development of attachment and well-being therefore offers a promising avenue for understanding not only average differences between single and partnered individuals but also why some single people thrive whereas others struggle. For other consistently single people, more insecure global and friend-specific attachment is characterized by rejection, conflict, or instability in social

relationships and likely co-occurs with adverse well-being trajectories. As described above, relatively insecure global attachment selects young adults into singlehood; attachment is subsequently less likely to be challenged by the socialization effects of committed partnership formation, allowing downstream consequences of insecure attachment for well-being to persist over time. Within the group of consistently single people, this would be reflected in relatively stable but low levels of global attachment as well as negative level-slope associations of attachment insecurity and well-being (i.e., higher initial attachment insecurity being linked to more adverse life satisfaction change).

Alternatively, long-term single people may gradually develop more maladaptive attachment orientations the longer they remain single. Such changes could emerge through the absence of attachment-relevant romantic experiences (Arriaga et al., 2018) or through repeated experiences associated with occupying a non-normative relationship status during a developmental period in which many peers form long-term partnerships (Arnett, 2000) and perhaps already start families of their own (Heckhausen et al., 2010; Wrosch & Heckhausen, 1999). Over time, these experiences may reinforce expectations of rejection, unavailability, or exclusion, particularly among individuals already vulnerable to attachment insecurity. This process is likely more pronounced for romantic and global attachment compared to friend- and parent-specific attachment, although empirical evidence disentangling these effects in single people is so far absent. Subsequently, maladaptive co-development of attachment and well-being would therefore contribute to the well-being disparities found in prior studies (e.g., Gonzalez Avilés et al., 2021; Krämer et al., 2026). Although this possibility appears somewhat inconsistent with evidence suggesting relative attachment stability in stable relational contexts, existing research provides little guidance regarding the developmental trajectories of attachment in long-term singlehood specifically (cf. Traut, Gander, Weidmann, et al., 2025), and regarding the different working models of attachment. Therefore, for co-development, we focus on single individuals' global and friend-specific attachment in confirmatory hypotheses and explore partner- and parent specific attachment without directed hypotheses.

The Present Study

Here, we use data from the yourPersonality Project (Fraley et al., 2021), a 3-year longitudinal study with approximately monthly assessments of attachment orientation (based on the full ECR-RS), life satisfaction, relationship status, and life events. Bringing together

theoretical and empirical perspectives on attachment, singlehood, and well-being, we aim to investigate the following research questions:

- RQ1: Do differences in attachment models among single people at baseline predict whether they stay single consistently or eventually find a partner?
- RQ2: How do consistently single people's attachment models change with increasing singlehood duration, compared to consistently and intermittently partnered individuals?
- RQ3: How do consistently single people's attachment models co-develop with life satisfaction with increasing singlehood duration?

We will examine both global and relationship-specific attachment models (i.e., with romantic partners, best friends, mothers, and fathers). As indicators of attachment insecurity, we will examine attachment avoidance and anxiety separately. Consistent singlehood will be operationalized as being single at each available wave and also reporting no relationship formation events that occurred in between waves. For RQ1, the analysis sample will comprise people who were single at baseline, comparing those who stayed single consistently with those who later found a partner within the observation period. For RQ2 and RQ3, the main comparison group of partnered individuals (for confirmatory analyses) will comprise participants who were consistently partnered across all waves. In additional, exploratory analyses we will compare single people with individuals who were intermittently partnered across waves, involving transitions into and out of relationships. For participants with skipped waves to be included in the analysis samples, we require at least three consecutive waves of data.

Based on theoretical reasoning and previous empirical findings, we preregister the following competing hypotheses, representing our primary aims to address the research questions. As secondary aims, we preregister exploratory analyses which complement the primary aims of the corresponding research question.

RQ1: Prospective Selection Effects

We hypothesize that single people with less secure global and romantic attachment models at baseline will be more likely to remain single throughout the observation period. However, it is also possible that global and romantic attachment models at baseline do not predict whether single people remain single throughout the observation period. The analyses

proposed below will allow us to test these possibilities. We will examine friend- and parent-specific attachment models in analogous exploratory analyses.

RQ2: Changes in Attachment Models

We expect that, over time, consistently single people will experience less pronounced increases in global and romantic attachment security compared to consistently partnered individuals (controlling for baseline age). At the same time, it is also possible that consistently single people do not differ from consistently partnered individuals in their trajectories of global and romantic attachment security. The preregistered analyses are intended to tease these alternatives apart.

Additionally, we will examine friend- and parent-specific attachment models in analogous exploratory analyses. For all attachment models, we will also explore whether consistently single people experience differential changes in attachment compared to intermittently partnered individuals.

RQ3: Co-Development of Attachment and Life Satisfaction

Last, we hypothesize that single people who experience greater increases in global or friend-specific attachment security will also show greater increases in life satisfaction, and vice versa (positive co-development). Alternatively, changes in global or friend-specific attachment security may not be linked to changes in life satisfaction (no co-development). The analyses we preregister below test these possibilities.

We will examine romantic and parent-specific attachment models in analogous exploratory analyses. In addition, for all attachment models, we will explore co-development between attachment and life satisfaction in (1) consistently partnered individuals and (2) intermittently partnered individuals.

Method

Sample

The study sample is based on the *yourPersonality Project* which was approved by the Institutional Review Board at the University of Illinois at Urbana-Champaign (protocol number 17,382). The sample potentially includes up to 4,920 people who completed at least three assessment waves (for more information, see Fraley et al., 2021). Of those, 39% were single at baseline, 35% dating, and 25% married, although it is still unknown how many would be classified as consistently single, consistently partnered, or intermittently partnered across the

study period (for similar considerations, see also Dugan et al., 2024). The sample is predominantly female (82%) and on average 35 years old ($SD = 12$). At the time of this preregistration, the first author (who will be running analyses) has not yet accessed these data.

The study spans up to 3 years in total, with the average number of completed assessment waves reported as around 10 in a previous study (Fraley et al., 2021). Participants were sent email reminders for assessments approximately monthly and received between \$1 and \$8 as Amazon vouchers for each survey they completed.

Measures

For full documentation of all measures assessed in the study, see “Codebook - Variable Descriptions.txt” attached to this preregistration.

Attachment

At each wave, participants reported on their attachment orientations using the Experiences in Close Relationships—Relationships Structures questionnaire (ECR-RS; Fraley et al., 2011), with three items for attachment anxiety and six items for attachment avoidance. Participants answered these items first for global attachment (i.e., close relationships in general), and subsequently for attachment models specific to romantic partners, best friends, mothers, and fathers (the latter two being skipped if a parent was reported as dead). For romantic-partner attachment, currently single people were instructed to consider their representation of potential romantic relationships without regard to a specific partner.

The attachment items thus differed as to whether the item referred to a specific type of relationship and, if so, which type. For global attachment, participants were instructed: “Please read each of the following statements and rate the extent to which you believe each statement best describes your feelings about close relationships in general”. For friend-specific attachment, they read: “Please rate the extent to which you agree or disagree with these items with respect to your best friend”). An example item for attachment avoidance is “I don’t feel comfortable opening up to this person”. All items were rated on a 5-point Likert scale from 1 (*strongly disagree*) to 5 (*strongly agree*). To construct anxiety and avoidance scores within each of the domains, we will form mean scores by averaging items and reverse scoring items where necessary.

Life Satisfaction

The 5-item Satisfaction with Life Scale (Diener et al., 1985) was used at each wave to assess life satisfaction. We will form a mean scale score based on these items.

Singlehood and Partnership Status

Romantic relationship status was assessed at each wave with the question “Are you currently in an exclusive romantic (dating/marital) relationship?”, with response options “No”, “Yes, I am dating someone”, and “Yes, I am married”.

In addition, 25 life events were assessed with a multiple-choice checklist and the instructions “Please indicate which of the following events have taken place [*time of last survey*]. You may select more than one event.”. Here, we consider the events “I started dating someone new”, “I became engaged to get married”, “I got married”, “My partner and I broke up”, and “My partner and I divorced”.

For someone to be included in the group of consistently single people, they need to have consistently answered the relationship status question with “No” and also not have selected any of the relationship life events mentioned above. For coding singlehood duration, we will add up the time from the first to the current measurement wave for each single person. Note that this does not represent total singlehood duration but singlehood duration during study participation. For someone to be included in the group of consistently partnered individuals, they need to have consistently answered the relationship status question with either “Yes, I am dating someone”, or “Yes, I am married”, while also never selecting the events indication new relationship formation, break-up, or divorce. Finally, for someone to be included in the group of intermittently partnered individuals, they need to have answered the relationship status question with “Yes, I am dating someone”, or “Yes, I am married” at least once and with “No” at least once. No additional conditions due to life events are applied. Altogether, we require at least 3 consecutively answered survey waves for each participant to be considered in either group.

Analytical Strategy

For descriptive purposes, we will report means and standard deviations of the constructs of interest as well as zero-order correlations among them.

Prospective Selection Effects (RQ1)

In the sample of single people at baseline, we will run logistic regression models (with a logit link function) with consistent singlehood (0 = *no, initially single but later find partner*, 1 =

yes, consistently single) as the outcome and both attachment anxiety and avoidance as predictors, which will be z-standardized. We will also include age and gender as control variables. Thus, we will run joint models for anxiety and avoidance but separate models for global attachment and each domain of attachment (i.e., in total 5 models with different attachment domains).

Changes in Attachment Models (RQ2)

To trace mean-level change in attachment across time, we will run univariate latent growth curve models based on manifest mean scale scores. We will start with separate models in each analysis group (consistently single, consistently partnered, intermittently partnered) and then extend those to multi-group models comparing consistently single people to each of the partnered groups. Likelihood ratio tests will indicate whether a model constraining the latent slope parameters to be equal across groups fits the data better compared to a model freely estimating them. The linear growth variable will be time at assessment (i.e., months centered at study entry). We run separate models for attachment anxiety and avoidance and for global attachment and each domain (2 x 5 attachment outcomes for each group comparison). Missing data will be handled through FIML. To adjust for differences in the age structures of the respective groups, we will add baseline age as a time-invariant control variable in the latent growth models (i.e., regressing the latent intercept and slope factors on age).

Co-Development of Attachment and Life Satisfaction (RQ3)

To examine co-development, we will estimate bivariate latent growth curve models of attachment and life satisfaction within the group of consistently single people, as well as exploratorily in the each of the partnered groups. These will build progressively on the univariate models (RQ2), adding life satisfaction as the second construct. The main effects of interest are the cross-construct intercept-slope correlations and slope-slope correlations. Again, we will run separate models for the different attachment variables (2 x 5 models in each group).

Transparency and Openness

At the time of the preregistration, the author who will later wrangle the data and run analyses has not accessed the data yet. Other coauthors who are providing the data have been involved in previous publications with these data on changes in attachment after life events (Fraley et al., 2021), person characteristics as moderators of event-related change (Gillath et al., 2024), co-development of global and partner-specific attachment (Dugan et al., 2022), within-person fluctuations in partner-specific attachment (Dugan et al., 2024), life events and

personality traits (Dugan et al., 2023), changes in narcissism (Wetzel et al., 2026), and cross-sectional analyses of attachment and well-being (Vahedi et al., 2026). No previous papers with these data focused explicitly on consistently single people, although one study used single people as a comparison group for examining development of global attachment before and after relationship transitions (Traut et al., 2026).

We will use R and specifically the ‘tidyverse’ packages for data cleaning and visualization (Wickham et al., 2019) and ‘lavaan’ for structural equation modeling (Rosseel, 2012). We will use $\alpha = .05$ as our main inference criterion. Data cleaning and analysis scripts will be posted on the OSF alongside reproducible HTML reports. Cleaned data with the relevant variables to reproduce analyses will also be posted on the OSF repository.

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